

Fernando Martinez-Lopez

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EDUCATION

Ph.D. in Computer Science

Aug 2024 – May 2028 | New York, USA

Fordham University

Research: Artificial Intelligence (AI), Reinforcement Learning (RL), Machine Learning (ML), Game and Decision Theory.

Roles: Research & Teaching Assistant

Master in Data Science - MSc

Jan 2022 – May 2023 | New York, USA

Fordham University

GPA: 4.00 / 4.00

Relevant Courses: Cloud Computing, Big Data Computing, Blockchain Technology, Data Mining, Machine Learning, and Deep Learning.

Roles: Research & Teaching Assistant (CISC 5835 Algorithms for Data Science & CISC 6745 Data Visualization)

Master of Business Administration - MBA

Mar 2021 – Mar 2022 | Washington, D.C., USA

Quantic School of Business and Technology

Grade: Honors

B.S. in Business Engineering

Aug 2014 – Apr 2018 | Santo Domingo, DR

Santo Domingo Institute of Technology (INTEC)

GPA: 3.81 / 4.00 | Summa Cum Laude

INTEC Meritorious Scholarship (top 1% program)

INDUSTRY EXPERIENCE

Fordham-IBM Research Intern

May 2025 – Jul 2025 | New York, USA

Fordham University

Quantitative Modeling - AVP MUSA CCAR Capital Planning

Jul 2023 – Aug 2024 | New York, USA

MUFG Americas

- Led the development and implementation of the CCAR Capital Planning interconnected model suite framework, enhancing operational efficiency through automation and significantly reducing the risk of human errors.
- Redeveloped CCAR quantitative models to improve performance and deploy new desk strategies, while upholding robustness and business soundness.
- Conduct model validation, performance monitoring, and stress testing under severe input conditions, ensuring model resilience and robustness against extreme operational and market challenges.
- Performed exploratory variable research to identify and integrate novel predictors, enhancing the current model suite's performance.

Data Scientist

Sep 2021 – Jul 2023 | New York, USA

Smart Energy

- Developed end-to-end Machine Learning models generating actionable business insights, enhancing marketing campaigns for 10,000+ Warm Leads
- Hands-on data analyses, A/B experiments, and modeling over enormous datasets towards developing insights, recommendations, and presenting results to various stakeholders.

Data Scientist - Research and Analytics Office

Apr 2020 – Sep 2021 | Santo Domingo, DR

General Directorate of Internal Taxes

- Created, validated, tested, and tuned Machine Learning algorithms to detect tax noncompliance, manage risk behavior profiles, reduce tax gaps, and build advanced analytics engines.
- Produced analytics to provide stakeholders with data-driven insights for improved tax auditing (the action through which Tax Administrations seek to prevent taxpayers from incurring tax evasion or fraud) by aligning with defined business objectives.
- Analyzed data and provided insights and actionable data-driven recommendations to stakeholders.

Data Analyst - Registry Office

Sep 2018 – Apr 2020 | Santo Domingo, DR

General Directorate of Internal Taxes

- Implemented Digital Forms of the National Taxpayer Registry.
- Designed and developed the new engine of automated tax assignation model for the Taxpayers' Registry based on the economic activities and the behavior of the taxpayers.
- Gained knowledge and Hands-on experience in Python and usage of machine learning algorithms.

Project Analyst

Aug 2017 – Sep 2018 | Santo Domingo, DR

INABIMA

- Establishment of a Risk-based Project Management System.
- Social Impact evaluated by Statistic indicators implementation.

SELECTED PUBLICATIONS

Fernando Martinez, Tao Li, Yingdong Lu, Juntao Chen, "Stackelberg Coupling of Online Representation Learning and Reinforcement Learning," In The Fourteenth International Conference on Learning Representations (ICLR 2026).

Fernando Martinez, Tao Li, Yingdong Lu, Juntao Chen, "*In-Context Reinforcement Learning via Communicative World Models*," Submitted to The International Joint Conferences on Artificial Intelligence (IJCAI 2026).

Fernando Martinez-Lopez, Juntao Chen, Yingdong Lu, "*SPRIG: Stackelberg Perception-Reinforcement Learning with Internal Game Dynamics*," Presented in AAAI 2025 Workshop on Planning and Reinforcement Learning (PRL) - Bridging the Gap Between AI Planning and Reinforcement Learning.

Pei-Yuan Zhou, Amame Takeuchi, **Fernando Martinez-Lopez**, Malikeh Ehghaghi, Andrew K. C. Wong, En-Shiun Annie Lee, "Benchmarking Interpretability in Healthcare Using Pattern Discovery and Disentanglement," Published in MDPI - Bioengineering, 2025.

Yijun Zhao, Alexander Borelli, **Fernando Martinez**, Haoran Xue, Gary M. Weiss, "*Admissions in the age of AI: detecting AI-generated application materials in higher education*," Nature Publishing Group UK, Scientific Reports, 2024.

Fernando Martinez, Mariyam Mapkar, Ali Alfatemi, Mohamed Rahouti, Yufeng Xin, Kaiqi Xiong, Nasir Ghani, "Redefining Cyber Resilience with Dual-Space Prototypical Networks for DDoS Attack Detection," In 33rd International Conference on Computer Communications and Networks (ICCCN 2024).

Fernando Martinez, Yijun Zhao, "Integrating Multiple Visual Attention Mechanisms in Deep Neural Networks," IEEE 47th Annual Computers, Software, and Applications Conference (COMPSAC 2023).

Elizabeth S. Thrall, **Fernando Martinez Lopez**, Thomas J. Egg, Seung Eun Lee, Joshua Schrier, Yijun Zhao, "Rediscovering the Particle-in-a-Box: Machine Learning Regression Analysis for Hypothesis Generation in Physical Chemistry," Journal of Chemical Education, ACS Publications, 2023.

SKILLS

Programming Languages — Python, R, SQL, Bash, C++, MongoDB, InfluxDB, Solidity, **Python** — Core, Debugging, Sklearn, NumPy, Pandas, Matplotlib, Seaborn, BeautifulSoup, **Data Science, Big Data & Machine Learning** — PyTorch, TensorFlow, Jax, PySpark, Snowflake, Deep Learning, LLMs, Computer Vision, **Cloud Computing** — Google Cloud Platform (GCP), Azure, Microsoft Fabric, Docker, Kubernetes, **BI** — Power BI, Tableau, SSIS, ETL, **Microsoft Office** — Excel, VBA, PPT, Word, Outlook

HONORS & AWARDS

Fordham-IBM Research Fellowship IBM	2025
AI Summer Research Fellowship GSAS Fordham University	2025
GSAS Graduate Research Assistanship Award GSAS, Fordham University	2022
Honors Quantic School of Business and Technology	2021
Full International Scholarship (Fordham University) Ministry of Higher Education, Science, and Technology of the Dominican Republic	2021
Full Scholarship Quantic School of Business and Technology	2021
Summa Cum Laude Santo Domingo Institute of Technology	2018
Meritorious Scholarship Santo Domingo Institute of Technology Awarded for top 1% program	2014

VOLUNTEERING

ICML 2026 Conference Reviewer	
ACC 2026 Conference Reviewer	
AAAI 2026 Conference  Reviewer	
INFOCOM 2025 - NGResNet  Reviewer	
TECHO  Volunteer	Santo Domingo, Dominican Republic